

# Graphing Functions and Transformations with GeoGebra

GeoGebra Graphing Calculator · GeoGebra

|            |             |                                   |
|------------|-------------|-----------------------------------|
| Years 9-10 | ~30 minutes | Mathematics · Algebra & Functions |
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**NOTICE — AI-generated worksheet**

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.geogebra.org/graphing>  
Curriculum links: AC9M10A05 · AC9M8A02 · exponential functions

## Learning intentions

- I can use GeoGebra to graph linear, quadratic and exponential functions from their equations.
- I can predict how changing a parameter (slope, intercept or coefficient) will transform a graph before testing it.
- I can use sliders to test and refine a conjecture about a family of functions.

## Getting started

Open the GeoGebra Graphing Calculator. Type  $y = x$  into the input bar to plot your first line, then click the small circle next to a number in the equation to turn it into a slider. Keep the input bar visible throughout, as you will be typing several equations.

## Tasks

1. Type  $y = 2x + 1$  into the input bar. Without changing anything yet, predict what will happen to the line if you change the 2 to a 5. Then test it — were you right?

2. Turn the '+1' in your equation into a slider and drag it between -5 and 5. Describe, in one or two sentences, what this number controls on the graph.

3. Graph four different linear equations of your choice and record how the slope value affects the steepness and direction of each line.

| Equation | Slope (m) | Effect on steepness/direction |
|----------|-----------|-------------------------------|
|          |           |                               |
|          |           |                               |
|          |           |                               |
|          |           |                               |

4. Type  $y = x^2$  into the input bar, then graph  $y = x^2 + 3$  and  $y = (x - 2)^2$  in the same window. Compare the three graphs and explain, in your own words, what adding a number outside versus inside the brackets does to a quadratic graph.

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5. Graph  $y = 2^x$ . Use the graph to estimate the value of  $x$  when  $y = 8$ . Check your estimate by evaluating  $2^x$  directly.

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6. Choose one family of functions (linear, quadratic or exponential). Write a general rule, in words, for how changing one parameter transforms the graph. Use at least one specific example from GeoGebra to support your rule.

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7. A phone plan costs \$20 plus \$0.50 per gigabyte of data used. Write this as a linear equation and graph it in GeoGebra. Use the graph to find how many gigabytes you can use for \$35.

Working:

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### ***Extension (optional)***

Sketch, on paper first, what you think  $y = -0.5(x + 3)^2 - 2$  will look like, including its turning point and direction. Then graph it in GeoGebra to check your prediction and describe how close you were.

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